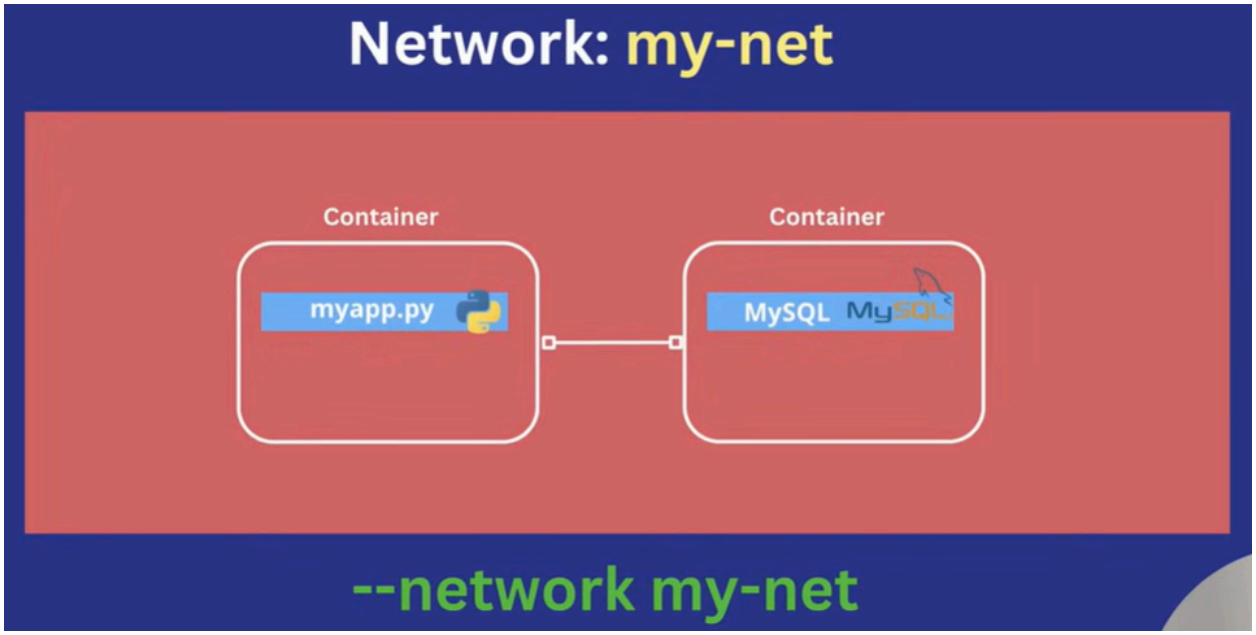
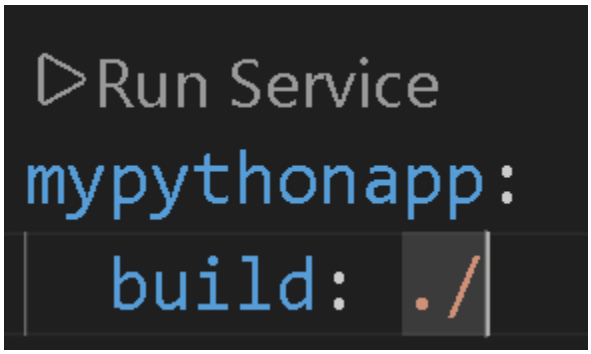
Docker Compose with Multiple Containers



MySQL is convered in docker-compose, how to take both of them and combine them together.
Docker-compose cant replace dockerfile,



./: meaning in current directory.

Code in docker-compose.yml:

```
services:
  mysqldb:
    image: "mysql:latest"
    environment:
      - MYSQL_ROOT_PASSWORD="enter your mysql password here…"
      - MYSQL_DATABASE="employee_db"
    container_name: "mysqldb"

  mypythonapp:
    build: ./
```

```
PS C:\Users\kiran\OneDrive\Desktop\Python Code> docker-compose up
[+] up 3/3
 ✓ Network pythoncode_default         Created
 ✓ Container pythoncode-mypythonapp-1 Created
 ✓ Container mysqldb                  Created
Attaching to mysqldb, mypythonapp-1
mysqldb  | 2026-03-24 10:14:52+00:00 [Note] [Entrypoint]: Entrypoi
```

In new terminal:

```
PS C:\Users\kiran\OneDrive\Desktop\Python Code> docker ps -a
CONTAINER ID   IMAGE                      COMMAND
2f5e625a4ddb   mysql:latest               "docker-entrypoint.s…"
6ba5cc0c18c6   pythoncode-mypythonapp     "python PyCode.py"
app-1
PS C:\Users\kiran\OneDrive\Desktop\Python Code>
```

```
docker ps -a
                 CREATED          STATUS                     PORTS                 NAMES
-entrypoint.s…"  30 seconds ago   Up 29 seconds              3306/tcp, 33060/tcp   mysqldb
PyCode.py"       30 seconds ago   Exited (0) 25 seconds ago                        pythoncode-mypython
```

```yaml
Run All Services
services:
  Run Service
  mysqldb_new:
    image: "mysql:latest"
    environment:
      - MYSQL_ROOT_PASSWORD="[redacted]"
      - MYSQL_DATABASE="employee_db"
    container_name: "mysqldb_new"

  Run Service
  mypythonapp:
    build: ./
    container_name: mypyapp
    depends_on:
      - 'mysqldb_new'
```

Until mysqldb_new is not up, hold the mypythonapp.

```
PS C:\Users\kiran\OneDrive\Desktop\Python Code> docker-compose up
[+] up 3/3
 ✓ Network pythoncode_default Created
 ✓ Container mysqldb_new      Created
 ✓ Container mypyapp          Created
Attaching to mypyapp, mysqldb_new
mysqldb_new  | 2026-03-24 12:27:59+00:00 [Note] [Entrypoint]: Entr
```

Initially python tries to be up

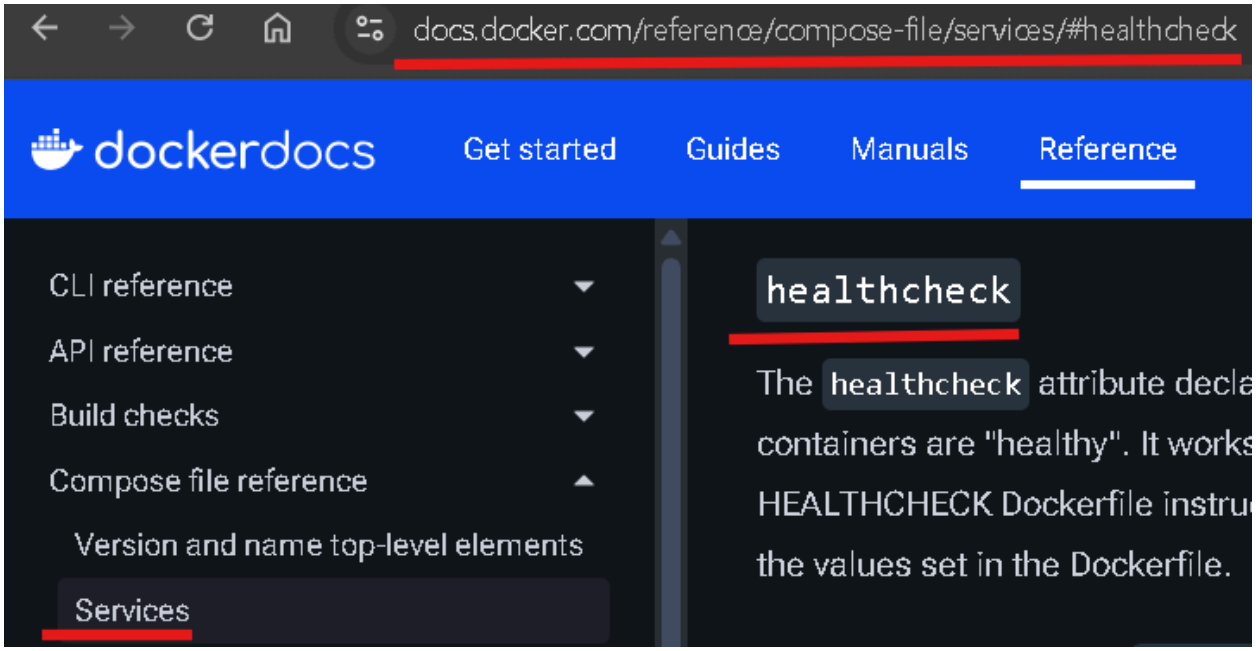
```
mysqldb_new  | 2026-03-24 12:28:01+00:00 [Note] [Entrypoint]: Switching to dedicated user 'mysql'
mysqldb_new  | 2026-03-24 12:28:01+00:00 [Note] [Entrypoint]: Entrypoint script for MySQL Server 9.6.0-1.el9 started.
mypyapp      | ✗ Database connection failed: (2003, "Can't connect to MySQL server on 'mysqldb_new' ([Errno 111] Connection refused)")
mysqldb_new  | 2026-03-24 12:28:01+00:00 [Note] [Entrypoint]: Initializing database files
mysqldb_new  | 2026-03-24T12:28:01.985139Z 0 [System] [MY-015017] [Server] MySQL Server Initialization - start.
mysqldb_new  | 2026-03-24T12:28:01.986235Z 0 [System] [MY-013169] [Server] /usr/sbin/mysqld (mysqld 9.6.0) initializing of server in pro
s 80
mysqldb_new  | 2026-03-24T12:28:02.016302Z 1 [System] [MY-013576] [InnoDB] InnoDB initialization has started.
mypyapp exited with code 0
mysqldb_new  | 2026-03-24T12:28:02.331408Z 1 [System] [MY-013577] [InnoDB] InnoDB initialization has ended.
mysqldb_new  | 2026-03-24T12:28:03.538388Z 6 [Warning] [MY-010453] [Server] root@localhost is created with an empty password ! Please co
```

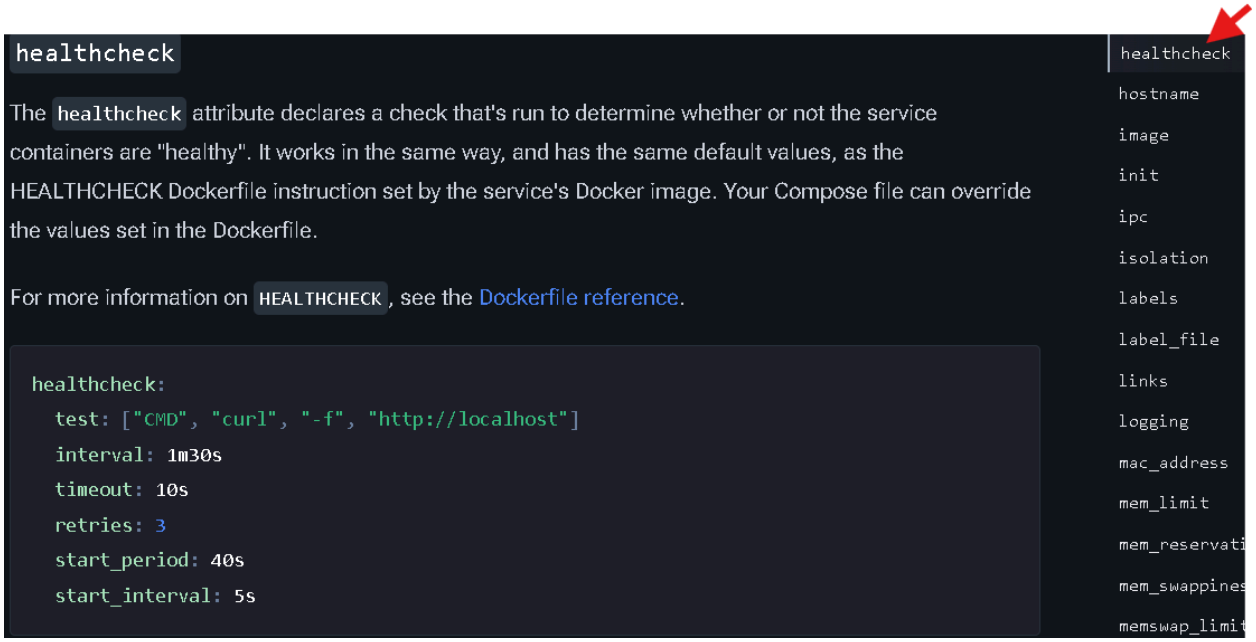
Since db is not fully up, py file can't connect to db.

```
mysqldb_new  | 2026-03-24T12:28:09.344347Z 0 [System] [MY-011323] [Server] X Plugin ready for connections.
/run/mysqld/mysqlx.sock

v View in Docker Desktop   o View Config   w Enable Watch   d Detach
```

At the end its ready for connection.
since , we have provided python depends on db, but why it still ran in the first place,





Added in compose file:

```
environment:
  - MYSQL_ROOT_PASSWORD="ghat123@gmail.com"
  - MYSQL_DATABASE="employee_db"
container_name: "mysqldb_new"
healthcheck:
  test: ["CMD", "mysqladmin", "ping", "-h", 'localhost']
  timeout: 20s
  retries: 10
```

`condition` : Sets the condition under which dependency is considered satisfied

- `service_started` : An equivalent of the short syntax described previously
- `service_healthy` : Specifies that a dependency is expected to be "healthy" (as indicated by `healthcheck`) before starting a dependent service.
- `service_completed_successfully` : Specifies that a dependency is expected to run to successful completion before starting a dependent service.

container_name
credential_spec
depends_on
deploy
develop
device_cgroup_ru
devices

```
services:
  web:
    build: .
    depends_on:
      db:
        condition: service_healthy
        restart: true
      redis:
        condition: service_started
  redis:
    image: redis
  db:
    image: postgres:18
```

So added in compose file:

```yaml
mypythonapp:
  build: ./
  container_name: mypyapp
  depends_on:
    mysqldb_new:
      condition: service_healthy
```

```
PS C:\Users\kiran\OneDrive\Desktop\Python Code> docker-compose down
[+] down 3/3
 ✓ Container mypyapp          Removed
 ✓ Container mysqldb_new      Removed
 ✓ Network pythoncode_default Removed
PS C:\Users\kiran\OneDrive\Desktop\Python Code>
```

```
PS C:\Users\kiran\OneDrive\Desktop\Python Code> docker-compose up
[+] up 3/3
 ✓ Network pythoncode_default Created
 ✓ Container mysqldb_new      Created
 ✓ Container mypyapp          Created
Attaching to mypyapp, mysqldb_new Config   w Enable Watch   d Detac
Container mysqldb_new Waiting
mysqldb_new  | 2026-03-24 13:00:59+00:00 [Note] [Entrypoint]: Entry
mysqldb_new  | 2026-03-24 13:00:59+00:00 [Note] [Entrypoint]: Switc
mysqldb_new  | 2026-03-24 13:00:59+00:00 [Note] [Entrypoint]: Entry
mysqldb_new  | 2026-03-24 13:00:59+00:00 [Note] [Entrypoint]: Initi
```

```
/run/mysqld/mysqlx.sock
mysqldb_new  | 2026-03-24T13:01:07.175814Z 0 [System] [MY-010931] [Server] /usr/sbin/mysqld: ready for connections. Versi
/mysqld/mysqld.sock'  port: 3306  MySQL Community Server - GPL.
Container mysqldb_new Healthy
mypyapp      | ✗ Database connection failed: (1045, "Access denied for user 'root'@'172.21.0.3' (using password: YES)")
mypyapp exited with code 0
```

Issue with authentication.

environment

The `environment` attribute defines environment variables set in the container. `environment` can use either an array or a map. Any boolean values; true, false, yes, no, should be enclosed in quotes to ensure they are not converted to True or False by the YAML parser.

Environment variables can be declared by a single key (no value to equals sign). In this case Compose relies on you to resolve the value. If the value is not resolved, the variable is unset and is removed from the service container environment.

Map syntax:

```
environment:
  RACK_ENV: development
  SHOW: "true"
  USER_INPUT:
```

Array syntax:

```
environment:
  - RACK_ENV=development
  - SHOW=true
  - USER_INPUT
```


develop
device_cgroup
devices
dns
dns_opt
dns_search
domainname
driver_opts
entrypoint
env_file
environment
expose
extends
external_link
extra_hosts
gpus
group_add
healthcheck
hostname
image
init

In environment values dont have double quotes.

```
mysqldb_new:
  image: "mysql:latest"
  environment:
    - MYSQL_ROOT_PASSWORD=[redacted]
    - MYSQL_DATABASE=employee_db
```

Above image shows it environment vairable does not have double quotes.

```
PS C:\Users\kiran\OneDrive\Desktop\Python Code> docker-compose down
[+] down 3/3
 ✓ Container mypyapp          Removed
 ✓ Container mysqldb_new      Removed
 ✓ Network pythoncode_default Removed
PS C:\Users\kiran\OneDrive\Desktop\Python Code> docker-compose up
[+] up 3/3
 ✓ Network pythoncode_default Created
 ✓ Container mysqldb_new      Created
 ✓ Container mypyapp          Created
Attaching to mypyapp, mysqldb_new
Container mysqldb_new Waiting
mysqldb_new  | 2026-03-24 13:06:03+00:00 [Note] [Entrypoint]: Entryp
mysqldb_new  | 2026-03-24 13:06:03+00:00 [Note] [Entrypoint]: Switch
```

```
mysqldb_new  | 2026-03-24T13:06:12.428904Z 0 [System] [MY-010931] [Server] /usr/sbin/mysqld: ready for connections.
/mysqld/mysqld.sock'  port: 3306  MySQL Community Server - GPL.
Container mysqldb_new Healthy
mypyapp      | ✅ Database connected successfully!
mypyapp      |
mypyapp      |
mypyapp      | ===== Employee Management =====
mypyapp      | 1. Add Employee
mypyapp      | 2. Update Employee
mypyapp      | 3. Show Employees
mypyapp      | 4. Delete Employee
mypyapp      | 5. Quit
mypyapp      |
mypyapp      | Enter choice (1-5): ❌ Database connection failed: EOF when reading a line
mypyapp exited with code 0
```

```
Enter choice (1-5): ❌ Database connection failed: EOF when reading a line
with code 0
```

Must be in interactive mode.
Add these two:

```yaml
mypythonapp:
  build: ./
  container_name: mypyapp
  depends_on:
    mysqldb_new:
      condition: service_healthy
  stdin_open: true
  tty: true
```

```
/mysqld/mysqld.sock'  port: 3306  MySQL Community Server - GPL.
Container mysqldb_new Healthy
mypyapp      | ✅ Database connected successfully!
mypyapp      |
mypyapp      |
mypyapp      | ===== Employee Management =====
mypyapp      | 1. Add Employee
mypyapp      | 2. Update Employee
mypyapp      | 3. Show Employees
mypyapp      | 4. Delete Employee
mypyapp      | 5. Quit
mypyapp      |
```

It stuck at this point.
Docker-compose has two services:

PyCode.py docker-compose.yml

mpose.yml

```yaml
Run All Services
services:
    Run Service
  mysqldb_new:
    image: "mysql:la
    environment:
      - MYSQL_ROOT_
      - MYSQL_DATABA
    container_name:
    healthcheck:
      test: ["CMD",
      timeout: 20s
      retries: 10

    Run Service
  mypythonapp:
    build: ./
```

If possible you can run services one by one if possible:

Run mysqldb under detach mode:

```
PS C:\Users\kiran\OneDrive\Desktop\Python Code> docker-compose run -d mysqldb_new
[+]  1/1te 1/1
 ✓ Network pythoncode_default Created
Container pythoncode-mysqldb_new-run-c3d0c4fdc3d9 Creating
Container pythoncode-mysqldb_new-run-c3d0c4fdc3d9 Created
f7a826b76917be9f97c27e4737ad475310526e24d21e339842de9e518578721a
PS C:\Users\kiran\OneDrive\Desktop\Python Code>
```

```
PS C:\Users\kiran\OneDrive\Desktop\Python Code> docker ps -a
CONTAINER ID   IMAGE          COMMAND                  CREATED
f7a826b76917   mysql:latest   "docker-entrypoint.s…"   40 seconds ago
c4fdc3d9
```

```
docker ps -a
       CREATED          STATUS                    PORTS                 NAMES
t.s…"  40 seconds ago   Up 40 seconds (healthy)   3306/tcp, 33060/tcp   pythoncode-mysqldb_new-run-c3d0
```

It is in running state.

Now let's run the python app without detach mode, we have enabled interactive terminal:

```
mypythonapp:
  build: ./
  container_name: mypyapp
  depends_on:
    mysqldb_new:
      condition: service_healthy
  stdin_open: true
  tty: true
```

Little patience is required: to get result:

```
PS C:\Users\kiran\OneDrive\Desktop\Python Code> docker-compose run mypythonapp
[+]  1/1t 1/11
 ✓ Container mysqldb_new Started
Container mysqldb_new Waiting
Container mysqldb_new Healthy
Container pythoncode-mypythonapp-run-187f554517ac Creating
Container pythoncode-mypythonapp-run-187f554517ac Created
✅ Database connected successfully!


===== Employee Management =====
1. Add Employee
2. Update Employee
3. Show Employees
4. Delete Employee
5. Quit

Enter choice (1-5): 
```

```
Enter choice (1-5): 3

--- View Employees ---
Show all? (y/n): y
⚠️No employees found.


===== Employee Management =====
1. Add Employee
2. Update Employee
3. Show Employees
4. Delete Employee
5. Quit

Enter choice (1-5): 1

--- Add Employee ---
Enter Name (required): name1 name2
Enter Phone (required): 24234
Enter Email (optional): 234@fdfa.com
Enter Salary (optional): 24234
Enter Aptitude Score (0-10) (required): 5.24234
✅ Employee added successfully!


===== Employee Management =====
1. Add Employee
2. Update Employee
3. Show Employees
4. Delete Employee
5. Quit

Enter choice (1-5): 1

--- Add Employee ---
Enter Name (required): name2 mane21
Enter Phone (required): 23234
```

```
Enter Email (optional): 2342@gmail.com
Enter Salary (optional): 23434
Enter Aptitude Score (0-10) (required): 7.2342
✅ Employee added successfully!


===== Employee Management =====
1. Add Employee
2. Update Employee
3. Show Employees
4. Delete Employee
5. Quit

Enter choice (1-5): 3

--- View Employees ---
Show all? (y/n): y

--- Employee Records ---
(1, 'name1 name2', '234@fdfa.com', '24234', 24234.0, 5.24234)
(2, 'name2 mane2l', '2342@gmail.com', '23234', 23434.0, 7.2342)


===== Employee Management =====
1. Add Employee
2. Update Employee
3. Show Employees
4. Delete Employee
5. Quit

Enter choice (1-5):
```